

Autodesk® Scaleform®

AMP

Scaleform Analyzer for Memory and Performance AMP

Alex Mantzaris
3.02
2012 6 15

Copyright Notice

Autodesk® Scaleform® 4.2

© 2012 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AUGI, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo) Built with ObjectARX (design/logo), Burn, Buzzsaw, CAiCE, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, DesignServer, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWFx, DXF, Ecotect, Evolver, Exposure, Extending the Design Team, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, Freewheel, GDX Driver, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-omatic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, RiverCAD, Robot, Scaleform, Scaleform Gfx, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Sparks, SteeringWheels, Stitcher, Stone, StormNET, Tinkerbox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

Autodesk Scaleform:

	AMP
	Autodesk Scaleform Corporation
	6305 Ivy Lane, Suite 310
	Greenbelt, MD 20770, USA
	www.scaleform.com
Email	info@scaleform.com
	+1 (301) 446-3200
Fax	+1 (301) 446-3199

目次

1	はじめに	1
1.1	AMP	2
1.2	AMP	3
2	AMP の使い方	10
2.1		10
2.2		11
2.3		12
2.4		12
2.5		17
2.6		18
3	プラットフォームとインテグレーションに関する注意.....	20
4	ビルドに関する注意	21
5	AMP サポートの追加	22
5.1	Debug	22
5.2		22
5.3		23
5.4		23
6	AMP 関連の FAQ	24
7	追加情報	25

1 はじめに

AMP™

Scaleform

CPU
Scaleform®

ActionScript

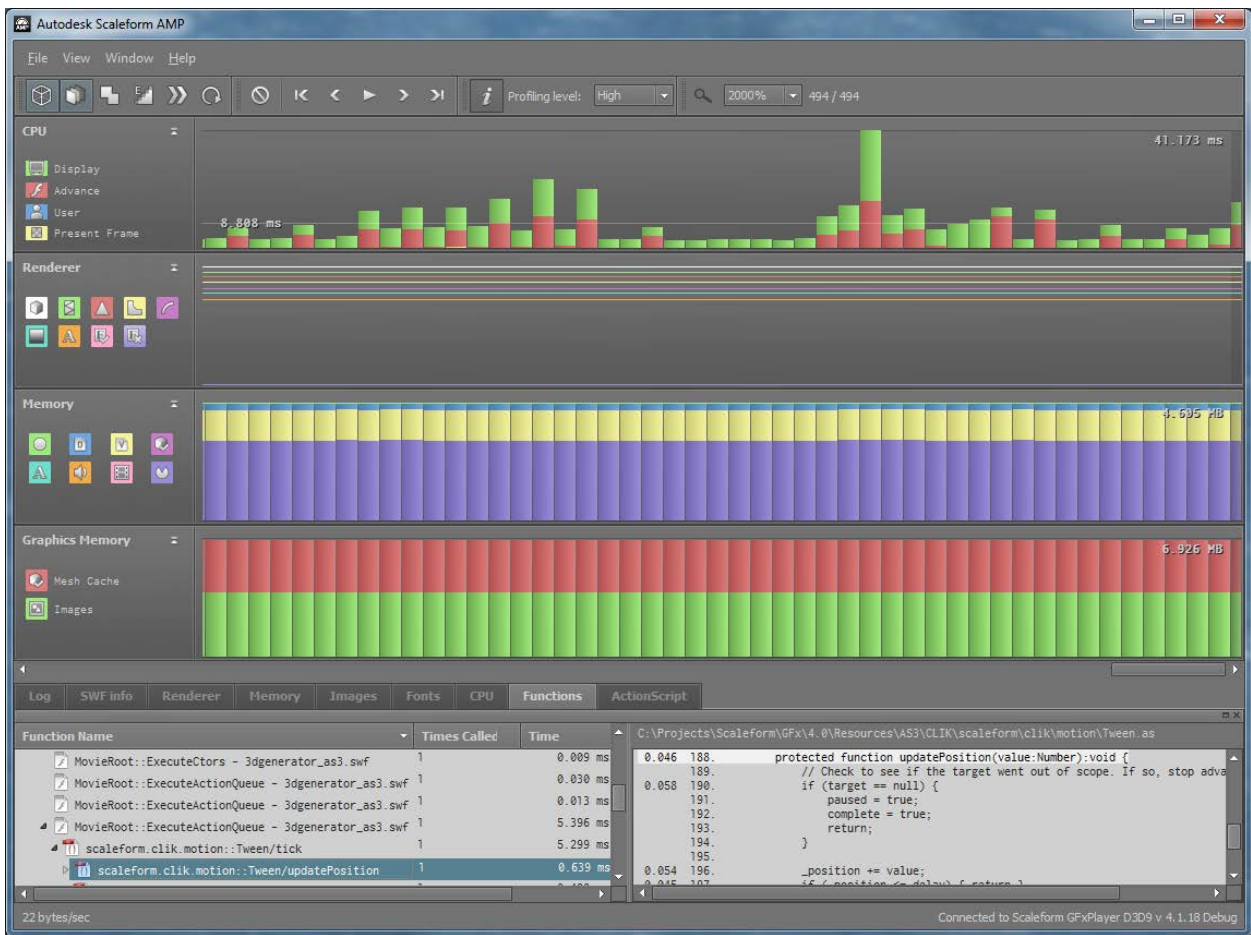


図 1: Scaleform AMP

Scaleform

AMP

AMP

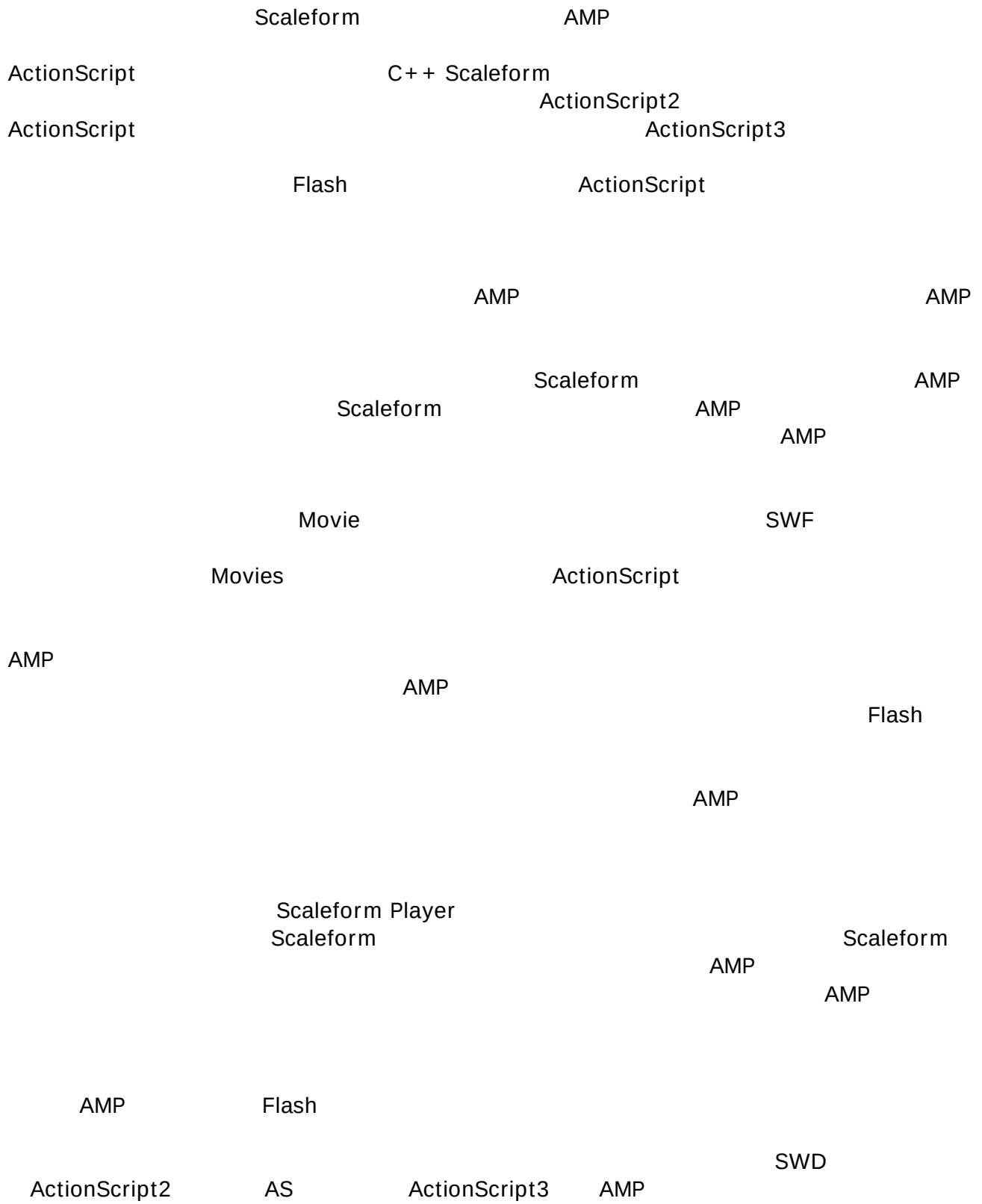
AMP

AMP

AMP

Scaleform 3.2

サーバー



クライアント

AMP

AMP

Scaleform

Scaleform CLIK

AMP

Flash

- **CPU グラフ:** Scaleform (Display) (PresentFrame) Direct Access API (User) Scaleform CPU
ActionScript (Advance)

- **レンダリンググラフ:** [Scaleform Best Practices Guide](#) Scaleform

Scaleform

- **メモリーグラフ:** Scaleform

- **グラフィックスメモリーのグラフ:** Scaleform
AMP

GPU

- **Log タブ:** Log
Scaleform

- **SWF info タブ:**

- **Renderer タブ:**

- **Memory タブ:** Scaleform Scaleform

AMP

i

- **Images タブ:**

Scaleform

- AMP
 - DXT
- Fonts タブ:**
 - SWF
- CPU タブ:**
 - User times
 - Advance
 - Display
- Functions タブ:**
 - ActionScript
 - C++
 - ActionScript VM
- ActionScript タブ:**
 - Functions
 - C++
- Source ウィンドウ:**
 - AS3
 - SWF
 - SA3
 - Functions
 - AS
 - ActionScript
 - AS2
 - High
 - Scaleform
 - AS2
 - 2
- Flash
 - SWF
 - AMP
 - AMP
 - SWF
 - AS3
- File**
 - Connection**
 - Disconnect :**
 - Debug Info Paths:**
 - AMP
 - AS2
 - SWD
 - AS3
 - AS
 - Add
 - Remove
 - Load Profile Frames**

- o **Save Profile Frames**
 - Load Profile Frames
- o **Dump Memory Report:**AMP
"AmpMemReport.txt"
- o **Exit :**
- **View menu:**
 - o **Graph Tooltips:** AMP
 - o **Time Units:**
 - o **Memory Units:**Memory Memory
- **Window menu:**
 - Log SWF Info Renderer Memory
 - Images Fonts CPU Functions ActionScript
 - Restore UI AMP
- **Help menu:**
 - 2
 - Scaleform.com User Guide
 - About AMP
- **Application Control Toolbar**
 - UI
 - o  **Wireframe**
 - o  **Overdraw mode**

AMP

0  **Batch profiling mode**

0  **Anti-aliasing**

Scaleform

EdgeAA

HW

FSAA

None) 3

0  **Fast-forward** Flash

SFW

FPS

0  **Restart**

- **Frame control ツールバー:**

○  **Clear**

AMP

○  **First Frame :**

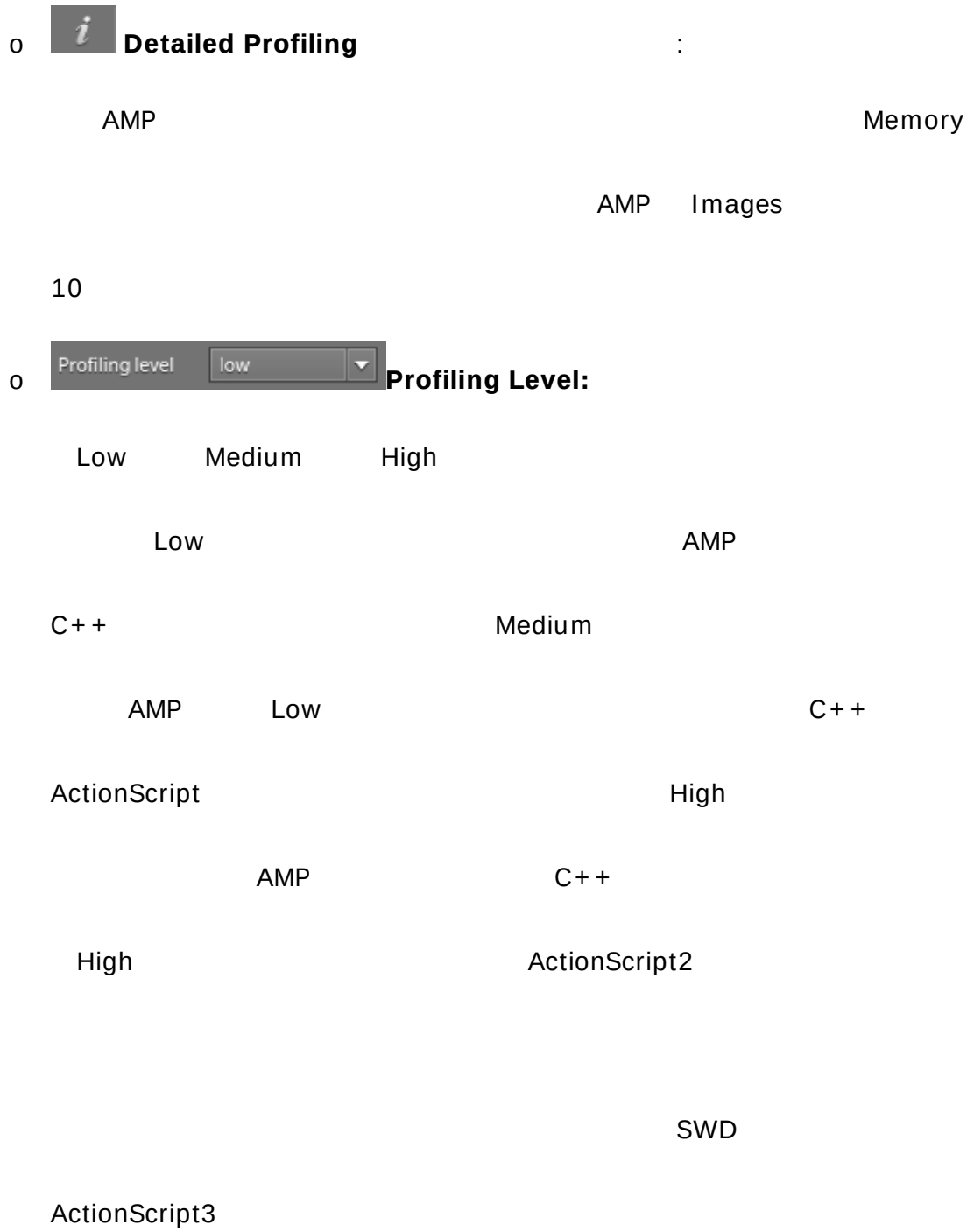
○  **Previous Frame :**

○  **Pause**

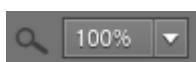
○  **Next Frame**

○  **Last Frame**

- **Profiling ツールバー:**



- **Zoom ツールバー:**



- **Status バー:**

Status

AMP

UI AMP
Reset UI

2 AMP の使い方

AMP
Bin/AMP GfxAmpClient.exe Scaleform

コネクションの確立

IP AMP AMP

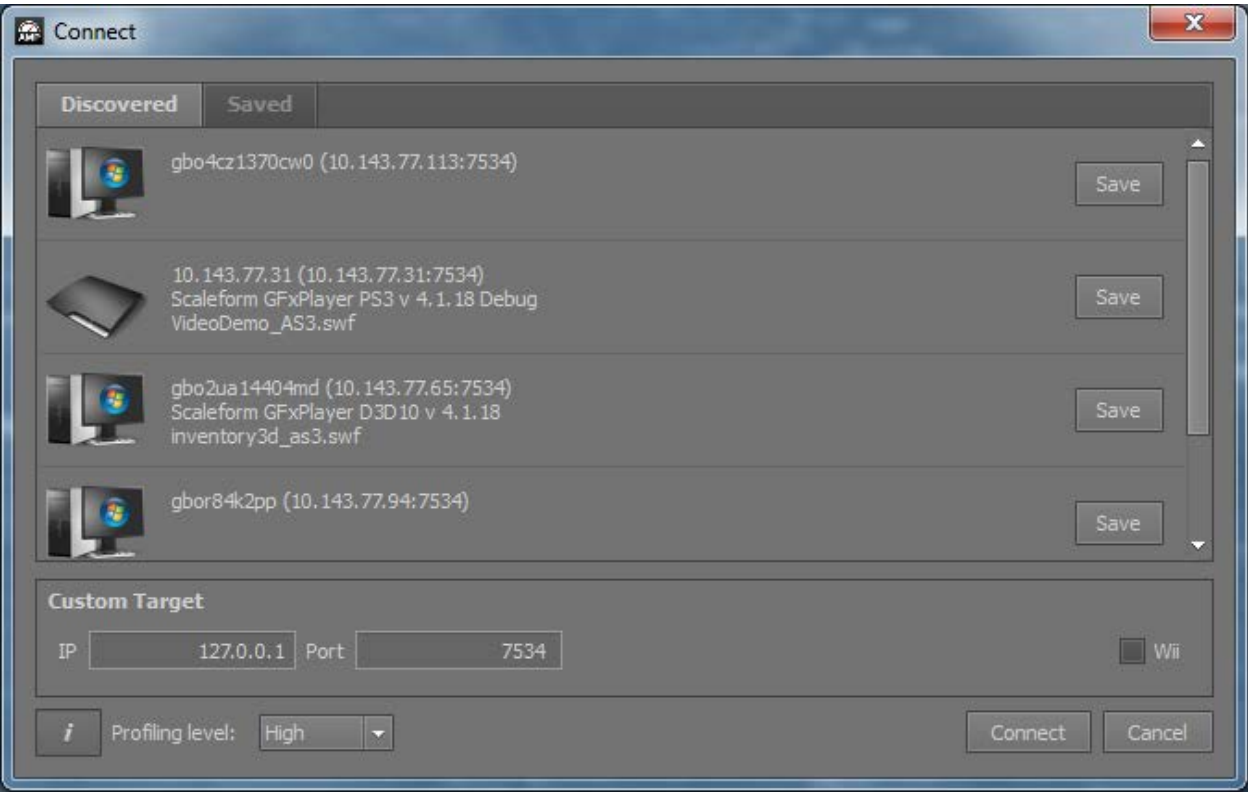
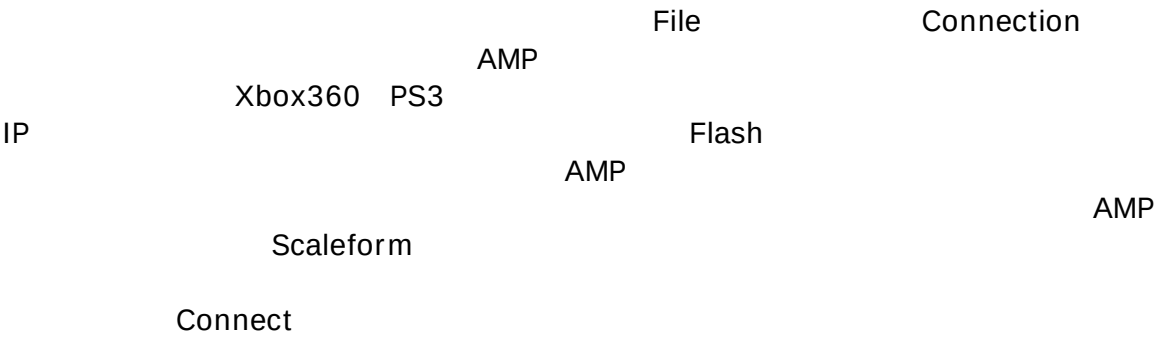


図 2 : AMP コネクション・ダイアログ



Wii NDEV NDEV USB COM Wii
NDEV Wii AMP IP wii

Advance Scaleform Display Action Script
Direct Access API User CPU



UI

ActionScript Function

ActionScript

11

図 4 : CPU Summary タブ

CPU

CPU Summary

レンダリング指標の解析

[Best Practices Guide](#)

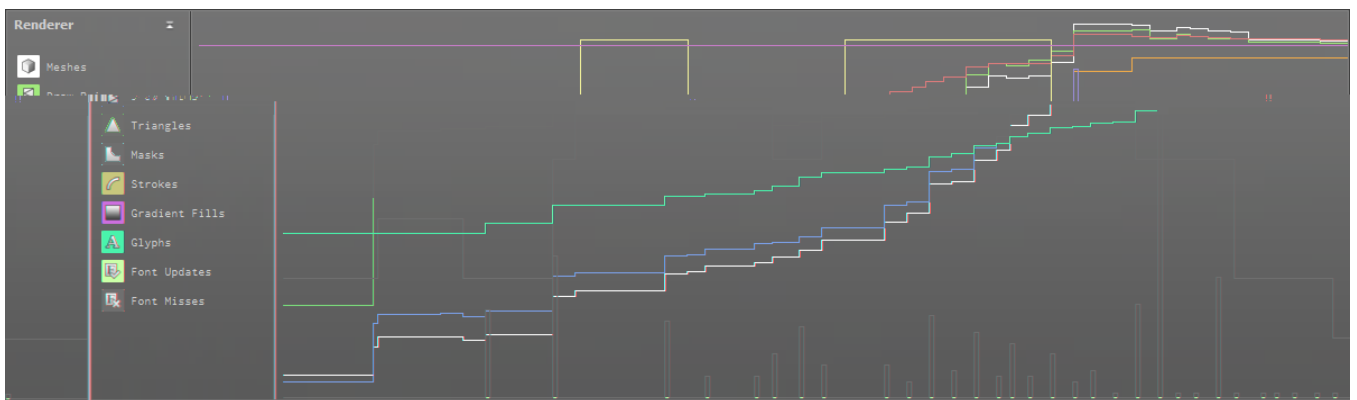


図 5: レンダリング・グラフ

Log	SWF info	Renderer	Memory
Meshes: 182			
Draw Primitives: 183			
Triangles: 5124			
Masks: 6			
Filters: 0			
Strokes: 21			
Gradient Fills: 0			
Rasterized Glyphs: 412			
Font Cache Updates: 0			
Font Textures: 1			
Font Cache Thrashing: 0			
Font Cache Fill: 28%			
Font Cache: 1048576 pixels			
Font Cache Used: 298294 pixels			
Font Failures: 0			
Font Cache Misses: 5			
Mesh Thrashing: 0			

図 6: Rendering Summary タブ

メモリ消費の解析

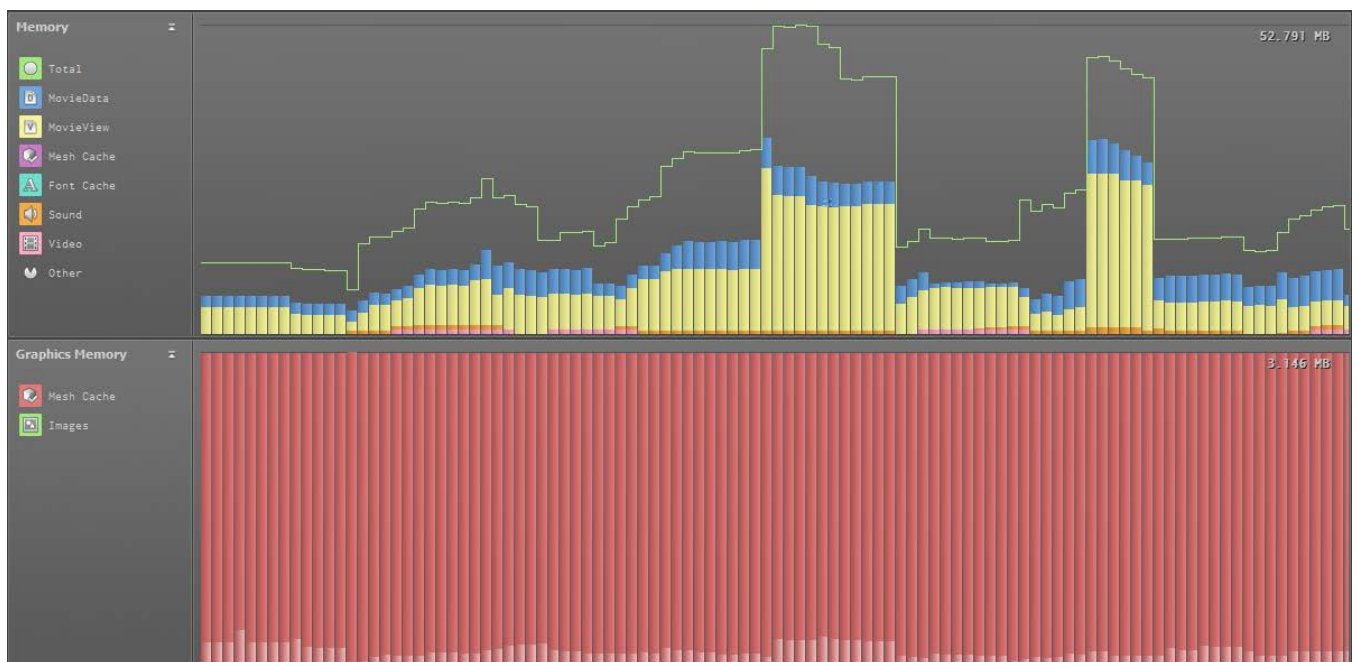


図 7: メモリ・グラフ

- **Total** Scaleform

- **MovieData** SWF GFX によって占有されて

- **Movie view** GfX::Movie

ActionScript

- **MeshCache**

- **Font Cache**

- **Sound**

- **Video**

- **Other** : Scaleform

-

Memory System Overview

“Used Space” Gfx

Debug Data Gfx
Memory

AMP

i

Scaleform

Scaleform

o **Renderer:**Scaleform

- **Buffers:**
- **RenderBatch:**
- **Primitive:**
- **Fill:**
- **Mesh:**
- **MeshBatch:**
- **Context:**
- **NodeData:**
- **TreeCache:**
- **TextureManager:**
- **MatrixPool:**
- **MatrixPoolHandles:**

- **Text:**
 - html
- **Font:**
- o **MovieDef:**
 - **CharDefs:**
 - **ShapeData:**
 - **Tags:**ActionScript
 - **Fonts:**
 - **Sounds:**
 - **ASBinaryData:**ActionScript
 - **MD_Other:** MovieDef
- o **MovieView:**
 - ActionScript
 - **MovieClip:**
 - **ActionScript:**ActionScript
 - **ASString:**ActionScript
 - **Text:**
 - **XML:**XML
 - **MV_Other:** Movie View
 - **VM:**ActionScript
 - **AS3 VM:**ActionScript 3
 - **CallFrame:**ActionScript
 - **VTable:**ActionScript
 - **SlotInfo:**ActionScript
 - ClassTraits
 - InstanceTraits
 - SlotInfo
 - ClassTraits
 - InstanceTraits
 - o **SlotInfoHash:**ActionScript
 - **ClassTraits:**ActionScript
 - **Class:**ActionScript
 - **InstanceTraits:**ActionScript
 - **Instance:**ActionScript
 - **AbcFile:**
 - o **AbcConstPool:**
 - AbcFile

- **VMABCFile:**AbcFiles

- **Tracer:**

- o **IME:**Input Method Editor (IME)

Log	SWF info	Renderer	Memory	Images
Memory		Size		
▲	Total Footprint	4849.402 KB		
▲	Used Space	4633.715 KB		
▷	Video Heaps	0.000 KB		
▷	Other Heaps	325.621 KB		
▲	Movie View Heaps	1226.094 KB		
▲	MovieView "3dgenerator_as3.swf"	1226.094 KB		
	String	0.828 KB		
▷	Renderer	12.672 KB		
▷	MovieView	1176.781 KB		
	General	35.812 KB		
▷	Movie Data Heaps	261.066 KB		
▲	Global Heap	2820.934 KB		
	Video	264.176 KB		
	String	20.633 KB		
	Sound	0.141 KB		
▷	Renderer	2310.914 KB		
▷	MovieView	8.039 KB		
▷	MovieDef	116.676 KB		
	Image	0.141 KB		
	IME	9.914 KB		
	General	90.301 KB		
▷	Debug Memory	8.000 KB		
▷	Unused Space	38.820 KB		
	Debug Data	176.867 KB		

图 8: Memory

Scaleform

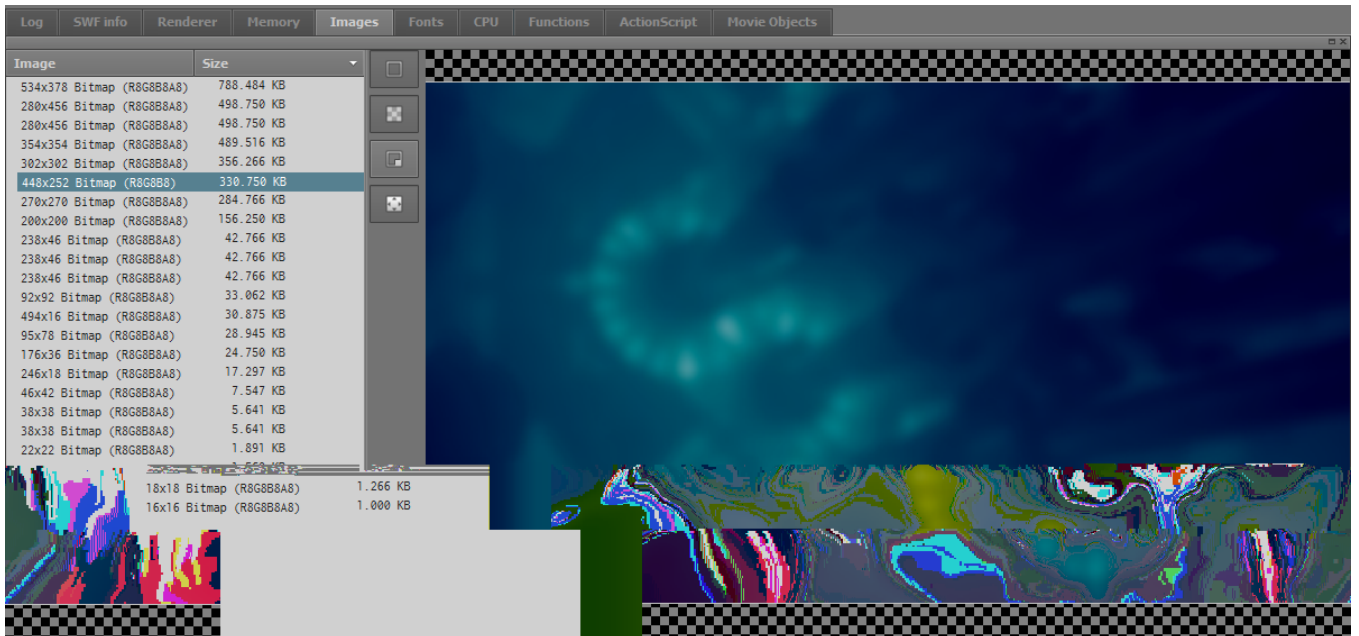


図 9: Images メモリタブ

Scaleform

Scaleform

Fonts



図 10: Fonts メモリタブ

SWF

SWF AMP MovieData CharDefs
Heaps Memory Summary

オーバードロー プロファイリングの使用

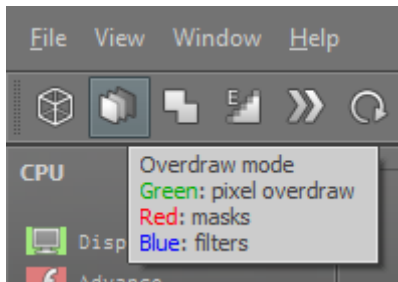


図 11: トグルボタン

16

8

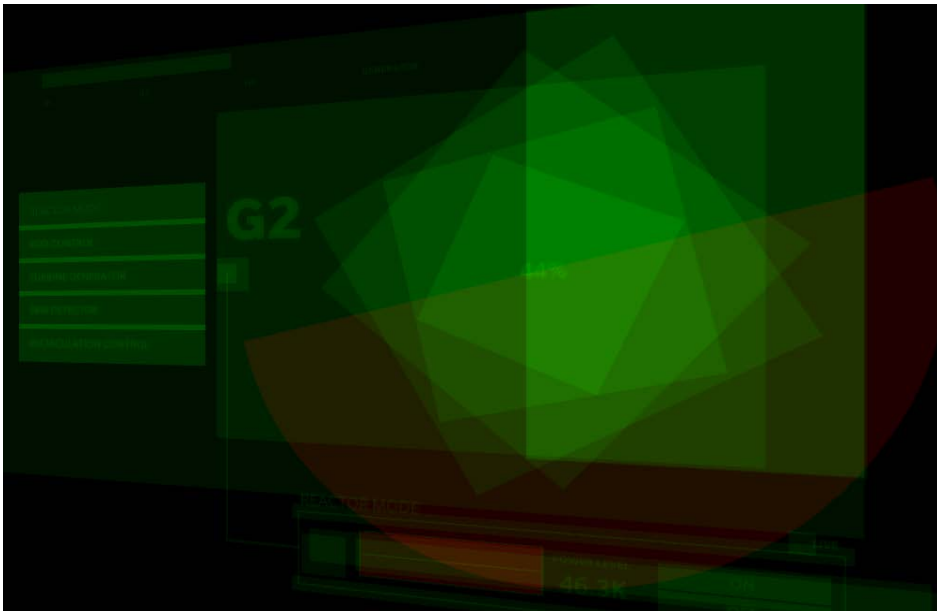


図 132: ハイライトレンダリング・モード

バッチ プロファイリングの使用

AMP

3 プラットフォームとインテグレーションに関する注意

- | | | | |
|--------|---------------------------------|----------------------------|-------------------|
| 1. AMP | GFx::System | Scaleform | GFx::System::Init |
| | AMP | Scaleform::System | |
| | GFx::AMP::Server::Init() | GFx::AMP::Server::Uninit() | |
| 2. AMP | Windows | Xbox360 | PS3 |
| | iPhone/iPad, 3DS, PS Vita, WiiU | Linux | Mac |
| | | | Wii, Android |

4 ビルドに関する注意

- AMP Scaleform
AMP GfxConfig.h GFX_AMP_SERVER
Scaleform Scaleform

```
AMP
Ptr<Gfx::AMP::ServerState> serverState = *SF_NEW Gfx::AMP::ServerState();
serverState->StateFlags &= Gfx::AMP::Amp_Disabled;
AmpServer::GetInstance().UpdateState(serverState);
```

- AMP

libnetctl Linux Windows Ws2_32.lib Xbox360 Xnet.lib PS3

- AMP AMP

AMP

```
AMP
AMP::Server::GetInstance().SetInitSocketLib(false)
AMP
```

- AMP Scaleform

```
AMP::Server::GetInstance().SetHeapLimit()
```

- AMP 7534

```
AMP::Server::GetInstance().SetListeningPort()
```

- AMP AMP

```
AMP::Server::GetInstance().SetConnectionWaitTime
```

5 AMP サポートの追加

Scaleform

AMP

AMP

Scaleform Player

ファイルの生成

AMP ActionScript 2 Flash

SWD

Flash CS3

CS4

Ctrl+Shift+Enter

SWD

SWD

SWF

AMP

ActionScript 3

SWD

SWF

AS

Flash Studio

SWF

アプリケーションのコントロール

AMP

AMP

app-control

- AMP::AppControlInterface app-control
AMP::MessageAppControl AMP
HandleAppRequest
- AMP::Server::GetInstance().SetAppControlCallback
- AMP::Server::GetInstance().SetAppControlCaps
AMP true
AMP::MessageAppControl

```
AMP::MessageAppControl caps;
caps.SetCurveToleranceDown(true);
caps.SetCurveToleranceUp(true);
caps.SetNextFont(true);
caps.SetRestartMovie(true);
caps.SetToggleAaMode(true);
caps.SetToggleAmpRecording(true);
caps.SetToggleFastForward(true);
caps.SetToggleInstructionProfile(true);
caps.SetToggleOverdraw(true);
caps.SetToggleBatch(true);
```

```
caps.SetToggleStrokeType(true);
caps.SetTogglePause(true);
caps.SetToggleWireframe(true);
AMP::Server::GetInstance().SetAppControlCaps(&caps);
```

コネクション・ステータス

AMP

```
AMP::Server::GetInstance().SetConnectedApp
```

AMP

- ServerStateType
 - AMP::Server::GetInstance().SetState
 - o Amp_RenderOverdraw
 - o Amp_App_Wireframe
 - o Amp_App_Paused
 - o Amp_App_FastForward
- AMP::Server::GetInstance().SetAaMode
- AMP::Server::GetInstance().SetStrokeType
- AMP::Server::GetInstance().SetCurrentLocale
- AMP::Server::GetInstance().SetCurveTolerance
- AMP::ServerState
 - AMP::Server::GetInstance().UpdateState

マーカー

C++

ActionScript

CPU

ActionScript

```
Amp.addMarker(1)
```

```
Amp.addMarker() .as
```

Resources/AS2/CLIK

Amp

C++

```
Gfx::MovieImpl:: AdvanceStats->AddMarker(1)
```

6 AMP 関連の FAQ

AMP

- ## 1. AMP を動作させるのにゲーム側で最低限必要な要件は何ですか。

```

AMP                                     GFX_AMP_SERVER
      GF                               AMP
Platform::RenderThread::drawFrame                                     AMP
      AmpServer::GetInstance().AdvanceFrame

```

2. AmpClient を使用してゲームに接続しようとする、ゲームのフレームレートが非常に低くなってしまう。AmpServer::AdvanceFrame() に多くの時間が取られています。これは通常の動作ですか。

AMP

AMP

AMP

i

3. ウィンドウズ、PS3、Xbox でゲーム実行中に、ゲームがそのディスカバリーウィンドウに AMP クライアントを発見した後でこれに接続できますが、統計は全て空白です。

AMP

Platform::RenderThread::drawFrame

AmpServer::GetInstance().AdvanceFrame

AMP

- #### 4. AMP がアップデートされるときにどうして 1 秒かそれ以上ポーズするのですか (AMP::Server::WaitForAmpConnection)。

AMP

1

Scaleform m

AMP

AMP

i

AMP

profile level

1MB

GfX::AMP::Server::SetHeapLimit

5. プロファイラー使用時にはどれほどのメモリーオーバーヘッドがありますか。

AMP

1MB

AmpServer::SetHeapLimit

AMP

GF

7 追加情報

AMP

- [AMP Use Cases](#)
- [Scaleform Reference](#)
- [AMP Forum](#)